Notes

Welcome and Introduction

Welcome to the Session

Hello everyone! Welcome to this mathematics session. I'm VNB Bhaskar Sharma, and I'm delighted to join you on your journey towards achieving a good JEE rank.

About the Instructor

I have 31 years of experience in the field of JEE coaching. I've worked with various coaching institutes across India and have taught all types of students, from top rankers to those who needed more support. I believe that mathematics is an ocean, and like you, I am still a student, constantly learning and growing. My motto is "Always keep learning," and my motive is "Seek knowledge and share knowledge."

Session Context: Month-End Batch

This session is part of the month-end batch, designed for students who have completed their 12th class and wish to make another attempt at JEE Advanced or JEE Mains. You will have two master teachers, and I am one of them.

Important Considerations for Droppers

Taking a drop year for JEE is a significant decision. It's crucial to have made this decision after careful deliberation with parents, friends, and family. Your friends might already be joining engineering colleges, and you might feel down at times, questioning your decision. However, once you've made the decision, it's important to stay focused and not fluctuate. This year will test your resolve, but with dedication and perseverance, you can achieve your goals. I wish you all the best for the coming year.

Introduction to the Course and Expectations

Welcome and Course Overview

This session marks the beginning of a comprehensive mathematics course for JEE Advanced, JEE Mains, and BITSAT aspirants. The instructor, Bhaskar Sharma, with 31 years of experience, will be covering Algebra and Coordinate Geometry.

Addressing Student Concerns and Mindset

- **Prior Knowledge:** Many students, especially 12th class completed or 13th droppers, might feel they already know certain topics. The instructor emphasizes the importance of patience, as even familiar topics might contain new or overlooked aspects.

- **Pace of Teaching:** The pace will be adjusted to cater to a diverse group of students. Feedback mechanisms will be available at the end of sessions to address concerns regarding speed or clarity.

Instructor's Approach

- **Theory First:** The primary focus will be on building a strong theoretical foundation before moving to problem-solving.
- **Experience-Driven:** The instructor will provide what is 'good for you' rather than just 'what you want', leveraging 31 years of teaching experience.
- **No Preconceptions:** The instructor will not form conclusions about students initially and requests the same from students regarding his teaching style.

Communication Guidelines

- **Chat Etiquette:** Students are requested to minimize non-essential chat during the lecture to ensure important doubts are not missed. The instructor's focus will be on teaching, not constantly monitoring the chat.
- **Feedback:** Specific feedback on pace, problem quantity, and understanding should be provided through the official feedback channels at the end of the session.

Course Structure and Resources

- **Faculty:** Two mathematics faculty members will be teaching; Amit sir will cover Calculus and other topics, while Bhaskar Sharma will handle Algebra and Coordinate Geometry.
- **Assignments and Homework:** Regular assignments and homework will be provided, along with a Telegram channel for further interaction.
- **PDF Notes:** PDFs of the sessions will be shared, so students do not need to rush to take notes during the lecture.

Expectations for Students

- **Utmost Attention and Focus:** Given the vast syllabus and the competitive nature of the exams, students are urged to give their full attention to avoid needing repetitions.
- **Prior Reading:** For this 'month-end batch', students are advised to do some prior reading before each session, as the teaching pace will be faster than a typical school environment.
- **Patience and Faith:** Students are asked to have patience and faith in the process, as the instructor aims to build that trust over time.

Today's Topic: Quadratic Equations

The session will begin with Quadratic Equations, starting from the basics. The instructor emphasizes that while JEE Mains might have straightforward problems from this topic, JEE

Notes

Welcome and Course Overview

This session serves as an introductory class. The primary focus for the initial four to six sessions will be on Quadratic Equations and the Theory of Equations. While this chapter is considered easy, students often have doubts, which the instructor encourages to be asked in appropriate contexts to maintain class pace.

Importance of Preparation

Students are advised to dedicate 1 to 1.5 hours to read the theory of quadratic equations and the theory of equations from their existing materials or recommended resources before the next session. This preparation is crucial for a smoother learning experience.

Addressing Student Queries and Concerns

The instructor emphasizes the importance of active listening during the lecture and encourages students to ask questions relevant to the current topic. There's a clear distinction made between general queries and doubts related to the ongoing discussion.

Homework for Day 1

The main homework for the first day is to read the theory of quadratic expressions and quadratic equations from any available material. This could include:

- Existing 11th and 12th-grade textbooks.
- Vedantu's Tattva books (highly recommended for Vedantu students).
- Searching "handbook math's handbook" on Google to find a publisher's handbook PDF for direct download and reading the relevant sections.

Recommended Study Materials

- **Theory:** Vedantu Tattva books are the first preference for Vedantu students. If Tattva books are unavailable, any other institute's theory books can be used.
- **Problems:** Tattva books, Previous Year Questions (PYQs), and assignments provided by the faculty (VDPP).

Accessing Resources and Support

Students are encouraged to connect with their Student Academic Mentor (SAM) to get access to the Telegram channel. Assignments and other important information will be shared on this channel. The instructor will also share VDPP materials and other resources via the Telegram channel.

There is no limit to the number of problems one should solve, but these core resources should be exhausted before seeking additional problems.

Final Remarks

The instructor acknowledges student feedback regarding the pace and style of the first class, assuring them that subsequent sessions will be different and more focused on the subject matter. Students are encouraged to communicate their understanding or lack thereof, as this helps the instructor adjust the teaching approach.

Introduction to Quadratic Equations

Definition of a Quadratic Equation

A quadratic equation is an equation of the form: $ax^2 + bx + c = 0$ where:

- a, b, and c are coefficients (real or complex numbers).
- $a \neq 0$ (the leading coefficient cannot be zero).

Coefficients

- a is called the leading coefficient.
- a, b, and c together are called the coefficients of the quadratic equation.

What is an Equation?

An equation represents a condition that is satisfied by a certain number of values (roots).

Roots of a Quadratic Equation

Definition of a Root

A value, say α (which can be a real or complex number), is called a root of a quadratic equation if, when substituted into the equation, it makes the equation equal to zero.

If $ax^2 + bx + c = 0$, then α is a root if $a\alpha^2 + b\alpha + c = 0$.

Number of Roots

For any equation, the number of roots is generally less than or equal to its degree. For a quadratic equation, the degree is 2 (the highest power of x). Therefore, a quadratic equation can have:

- Zero real roots
- One real root

exactly two roots (which can be real, complex, distinct, or repetitive).

Methods to Find Roots of a Quadratic Equation

There are two primary methods to find the roots of a quadratic equation:

1. Factorization Method

This method involves factorizing the quadratic expression $ax^2 + bx + c$ into two linear factors and then equating each factor to zero to find the roots.

2. Quadratic Formula (Siddha Acharya Method)

This is a general formula that can be used to find the roots of any quadratic equation. For an equation $ax^2 + bx + c = 0$, the roots are given by: $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

Discriminant

The term $b^2 - 4ac$ within the quadratic formula is called the **discriminant**, often denoted by Δ or D. It plays a crucial role in determining the nature of the roots.

Alternative Form of a Quadratic Expression

Any quadratic expression $ax^2 + bx + c$ can be rewritten in a perfect square form, which is useful for understanding the derivation of the quadratic formula and other properties: $ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + \frac{4ac-b^2}{4a}$ This form shows how the quadratic expression relates to a perfect square and a constant term, which is directly linked to the discriminant.

Introduction to Quadratic Equations and Roots

Definition of a Root

A root of an equation is a value that, when substituted into the equation, makes the equation equal to zero.

Quadratic Equation Standard Form

A standard quadratic equation is given by $ax^2 + bx + c = 0$, where $a \neq 0$.

Quadratic Formula

The roots of a quadratic equation $ax^2 + bx + c = 0$ are given by the quadratic formula (also known as the Sridharacharya method): $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$ Let the two roots be α and β:

- $\alpha = \frac{-b+\sqrt{b^2-4ac}}{2a}$

Notes

For a quadratic equation $ax^2 + bx + c = 0$ with roots α and β:

Sum of the Roots (S1)

This is referred to as the sum of the roots taking one at a time. $S_1 = \alpha + \beta = -\frac{b}{a}$

Product of the Roots (S2)

This is referred to as the sum of the roots taking two at a time. Since there are only two roots, this simplifies to their product. $S_2 = \alpha \cdot \beta = \frac{c}{a}$

Forming a Quadratic Equation from its Roots

If the roots of a quadratic equation are known as α and β, the equation can be formed using the following methods:

Method 1: Factor Form

If α and β are the roots, then $(x - \alpha)$ and $(x - \beta)$ are factors of the quadratic equation. Thus, the equation can be written as: $a(x - \alpha)(x - \beta) = 0$ where a is the leading coefficient.

Method 2: Using Sum and Product of Roots

Expanding the factor form, we get: $a(x^2 - (\alpha + \beta)x + \alpha\beta) = 0$ Substituting $S_1 = \alpha + \beta$ and $S_2 = \alpha\beta$, the quadratic equation can be expressed as: $a(x^2 - S_1 x + S_2) = 0$ If $a = 1$, the equation is simply: $x^2 - (\text{sum of roots})x + (\text{product of roots}) = 0$ $x^2 - S_1 x + S_2 = 0$

Context and Importance of Quadratic Equations

Subject Overview

This session is part of the algebra module, which also covers coordinate geometry. The instructor is VNB Bhaskar Sharma.

Relevance to Exams

- **JEE Mains:** Quadratic equations are considered relatively easy if the theoretical concepts are well understood.
- **JEE Advanced:** Questions on quadratic equations are often mixed with other topics, such as binomial theorem, and can extend to the broader 'Theory of Equations' which includes cubic equations and their nature. For example, a past JEE Advanced paper (2025, first paper, first question) involved a mix with binomial theorem and dealt with the nature of a cubic equation, highlighting the importance of understanding the 'nature of roots' concept.

Notes

- For a polynomial, the values that make the polynomial equal to zero are called **zeros**.
- For a quadratic equation, these values are commonly referred to as **roots**.
- Essentially, both terms refer to the same concept: the values of the variable for which the expression evaluates to zero. The terminology often depends on whether the expression is considered a general polynomial or specifically a quadratic equation.

Understanding S1 and S2 in Polynomials

Sum of Roots Notation

In the context of polynomials, especially for higher degrees, specific notations are used to represent sums of roots:

- **S1**: Represents the **sum of the roots taken one at a time**. If the roots of a polynomial are $\alpha_1, \alpha_2, \ldots, \alpha_n$, then $S1 = \alpha_1 + \alpha_2 + \cdots + \alpha_n$.
- **S2**: Represents the **sum of the products of the roots taken two at a time**. For roots $\alpha_1, \alpha_2, \ldots, \alpha_n$, $S2 = \alpha_1\alpha_2 + \alpha_1\alpha_3 + \cdots + \alpha_{n-1}\alpha_n$.

Important Clarification

It is crucial to understand that in higher mathematics, S2 does *not* mean the product of the roots. The product of the roots is a different concept, often denoted as Sn for an n-degree polynomial (the sum of products of roots taken n at a time).

Relations Between Roots and Coefficients

Difference of Roots

To find the difference of the roots, $\alpha - \beta$, we use the identity: $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$ We know that for a quadratic equation $ax^2 + bx + c = 0$, the sum of the roots is $\alpha + \beta = -b/a$ (denoted as S_1) and the product of the roots is $\alpha\beta = c/a$ (denoted as S_2). Substituting these into the identity: $(\alpha - \beta)^2 = (S_1)^2 - 4S_2$ This allows us to find the square of the difference of the roots. To find $|\alpha - \beta|$, we take the square root of this expression.

Sum of Squares of Roots

To find the sum of the squares of the roots, $\alpha^2 + \beta^2$, we use the identity: $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ Substituting S_1 and S_2: $\alpha^2 + \beta^2 = (S_1)^2 - 2S_2$

Difference of Squares of Roots

To find the difference of the squares of the roots, $|\alpha^2 - \beta^2|$, we use the identity: $|\alpha^2 - \beta^2| = |(\alpha + \beta)(\alpha - \beta)|$ $|\alpha^2 - \beta^2| = |S_1 \cdot (\alpha - \beta)|$ Here, $(\alpha - \beta)$ can be found from the formula derived for the

Notes

To find the sum of the cubes of the roots, $\alpha^3 + \beta^3$, we use the identity: $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$ Substituting S_1 and S_2: $\alpha^3 + \beta^3 = (S_1)^3 - 3S_2(S_1)$

Difference of Cubes of Roots

To find the difference of the cubes of the roots, $\alpha^3 - \beta^3$, we use the identity: $\alpha^3 - \beta^3 = (\alpha - \beta)^3 + 3\alpha\beta(\alpha - \beta)$ Here, $(\alpha - \beta)$ can be found from the formula derived for the difference of roots.

Reciprocal Sum of Squares of Roots

To find $1/\alpha^2 + 1/\beta^2$, we can simplify the expression: $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\beta^2+\alpha^2}{(\alpha\beta)^2}$ Using the formula for $\alpha^2 + \beta^2$ and $(\alpha\beta)^2$: $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{S_1^2-2S_2}{S_2^2}$

Other Important Relations

- $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$: $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{S_1^2-2S_2}{S_2}$

These algebraic identities are crucial for solving problems involving relations between roots. The most frequently used identities are those for $(\alpha + \beta)^2$, $(\alpha - \beta)^2$, and $\alpha^2 + \beta^2$. Understanding how to express these in terms of S_1 and S_2 is fundamental.

Formation of Quadratic Equation from Roots

General Form

If the roots of a quadratic equation are given as α and β, the equation can be formed using the formula:

$$x^2 - (\text{sum of the roots})x + (\text{product of the roots}) = 0$$

This can also be written as:

$$x^2 - S_1 x + S_2 = 0$$

Where:

- $S_1 = \alpha + \beta$ (sum of the roots)
- $S_2 = \alpha \cdot \beta$ (product of the roots)

Relation between Roots and Coefficients

For a quadratic equation $ax^2 + bx + c = 0$, with roots α and β:

Notes

Difference of Roots

Derivation of Difference of Roots

To find the difference between the roots, $\alpha - \beta$, we can use the algebraic identity:

$(a - b)^2 = (a + b)^2 - 4ab$

Applying this to the roots α and β:

$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

Substituting the sum and product of roots:

$(\alpha - \beta)^2 = S_1^2 - 4S_2$

Therefore, the absolute difference of the roots is:

$|\alpha - \beta| = \sqrt{S_1^2 - 4S_2}$

Alternatively, for the quadratic equation $ax^2 + bx + c = 0$, the difference of roots can be expressed as:

$|\alpha - \beta| = \frac{\sqrt{D}}{|a|}$

Where $D = b^2 - 4ac$ is the discriminant.

Symmetric Functions of Roots

Definition of Symmetric Function

A function of roots is considered symmetric if, upon interchanging any two roots (e.g., replacing α with β and β with α), the value of the function remains unchanged. For example, $\alpha + \beta$ is a symmetric function because $\beta + \alpha$ is the same. Similarly, $\alpha \cdot \beta$ is symmetric.

Common Symmetric Relations

It is crucial to understand how to derive these relations rather than memorizing the final formulas. These are based on fundamental algebraic identities.

1. **Sum of Squares:** $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = S_1^2 - 2S_2$
2. **Difference of Squares:** $\alpha^2 - \beta^2 = (\alpha + \beta)(\alpha - \beta)$ To calculate this, first find $(\alpha - \beta)$ using the method described above.
3. **Sum of Cubes:** $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = S_1^3 - 3S_2S_1$

6. **Sum of Fifth Powers (Rarely Used but Good to Know):** $\alpha^5 + \beta^5 = (\alpha^2 + \beta^2)(\alpha^3 + \beta^3) - \alpha^2\beta^2(\alpha + \beta)$ Substitute the expressions for sum of squares, sum of cubes, and sum/product of roots to evaluate this.

Transformations of Quadratic Equations

Introduction

When given a quadratic equation $Ax^2 + Bx + C = 0$ with roots α and β, we often need to find a new quadratic equation whose roots are related to α and β through some transformation. This process is called the transformation of equations. We will explore various common transformations and the corresponding changes to the original equation.

1. Roots Transformed by Addition/Subtraction

If the new roots are $\alpha + k$ and $\beta + k$ (i.e., a constant k is added to each root):

- **Rule:** In the original equation, replace x with $(x - k)$.
- **New Equation:** $A(x - k)^2 + B(x - k) + C = 0$

If the new roots are $\alpha - k$ and $\beta - k$ (i.e., a constant k is subtracted from each root):

- **Rule:** In the original equation, replace x with $(x + k)$.
- **New Equation:** $A(x + k)^2 + B(x + k) + C = 0$

2. Roots Transformed by Multiplication/Division

If the new roots are $k\alpha$ and $k\beta$ (i.e., each root is multiplied by a constant k):

- **Rule:** In the original equation, replace x with $\frac{x}{k}$.
- **New Equation:** $A\left(\frac{x}{k}\right)^2 + B\left(\frac{x}{k}\right) + C = 0$

If the new roots are $\frac{\alpha}{k}$ and $\frac{\beta}{k}$ (i.e., each root is divided by a constant k):

- **Rule:** In the original equation, replace x with kx.
- **New Equation:** $A(kx)^2 + B(kx) + C = 0$

3. Roots with Opposite Signs

If the new roots are $-\alpha$ and $-\beta$ (i.e., each root is multiplied by -1):

- **Rule:** In the original equation, replace x with $(-x)$.
- **New Equation:** $A(-x)^2 + B(-x) + C = 0 \implies Ax^2 - Bx + C = 0$

- **Rule:** In the original equation, replace x with $\frac{1}{x}$.
- **New Equation:** $A\left(\frac{1}{x}\right)^2 + B\left(\frac{1}{x}\right) + C = 0$
- **Simplification:** Multiply by x^2 to clear denominators: $A + Bx + Cx^2 = 0$, or $Cx^2 + Bx + A = 0$.

5. Combined Transformation: Reciprocal and Opposite Sign

If the new roots are $-\frac{1}{\alpha}$ and $-\frac{1}{\beta}$:

- **Rule:** In the original equation, replace x with $-\frac{1}{x}$.
- **New Equation:** $A\left(-\frac{1}{x}\right)^2 + B\left(-\frac{1}{x}\right) + C = 0$
- **Simplification:** $A\left(\frac{1}{x^2}\right) - \frac{B}{x} + C = 0 \implies A - Bx + Cx^2 = 0$, or $Cx^2 - Bx + A = 0$.

6. General Linear Transformation of Roots

If the new roots are $P\alpha + Q$ and $P\beta + Q$ (i.e., each root is multiplied by P and then Q is added):

- **Rule:** Let $y = Px + Q$. Then $x = \frac{y-Q}{P}$. In the original equation, replace x with $\frac{x-Q}{P}$.
- **New Equation:** $A\left(\frac{x-Q}{P}\right)^2 + B\left(\frac{x-Q}{P}\right) + C = 0$

7. Roots Raised to a Power

If the new roots are α^n and β^n:

- **Rule:** In the original equation, replace x with $x^{1/n}$.
- **New Equation:** $A(x^{1/n})^2 + B(x^{1/n}) + C = 0$

8. Roots with Fractional Power (e.g., Square Root)

If the new roots are $\alpha^{1/n}$ and $\beta^{1/n}$ (e.g., $\sqrt{\alpha}$ and $\sqrt{\beta}$):

- **Rule:** In the original equation, replace x with x^n.
- **New Equation:** $A(x^n)^2 + B(x^n) + C = 0$

These transformations are crucial for solving problems where the roots of a new equation are derived from the roots of a given equation.

Importance of Prior Reading and Active Learning

The Importance of Preparation

problems, and delving into more complex topics rather than re-teaching basic theory.

- **Why Prior Reading?**
  - It helps students grasp concepts more effectively during the lecture.
  - It allows the lecturer to cover a wider range of problems and advanced models.
  - It prevents the class from becoming a repetitive re-teaching of 11th and 12th-grade material.
  - It is crucial for achieving high ranks in competitive exams like JEE Mains, BITSAT, and Advanced.

Consequences of Not Preparing

Without prior reading, the lecturer might have to spend more time on basic theory, which could lead to:

- **Reduced Problem-Solving:** Fewer advanced problems and models will be covered.
- **Compromised Rank Potential:** The goal of achieving a high rank might not be met due to insufficient practice and exposure to diverse problem types.
- **Extended Course Duration:** The need to cover basic theory repeatedly could necessitate a longer course duration (e.g., 1.5 to 2 years instead of 1 year).

Lecturer's Commitment

The lecturer is committed to student success and ranks, not just ratings. While it's easier to re-teach everything for student satisfaction, it's not beneficial for their long-term academic goals. The lecturer will guide students on what to read for the next class, suggesting 15 minutes of reading from textbooks or online resources.

Transformations of Roots in Quadratic Equations

General Method for Root Transformations

This section outlines various transformations that can be applied to the roots of a quadratic equation to find a new equation. These methods are not limited to quadratic equations but can also be applied to higher-degree polynomials (3rd, 4th, 5th, 6th degree, etc.).

Let the original quadratic equation have roots α and β.

1. **Roots are $p\alpha + q$ and $p\beta + q$:**
   - If the new roots are of the form $p\alpha + q$ and $p\beta + q$, replace x in the old equation with $\frac{x-q}{p}$.

2. **Roots are $-\alpha$ and $-\beta$:**
    - Replace x in the old equation with $-x$.
3. **Roots are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$:**
    - Replace x in the old equation with $\frac{1}{x}$.
4. **Roots are α^n and β^n:**
    - Replace x in the old equation with $x^{\frac{1}{n}}$.
5. **Roots are $k\alpha$ and $k\beta$ (multiplying roots by k):**
    - Replace x in the old equation with $\frac{x}{k}$.
6. **Roots are $\alpha + k$ and $\beta + k$ (adding k to roots):**
    - Replace x in the old equation with $x - k$.
7. **Roots are $\frac{\alpha}{k}$ and $\frac{\beta}{k}$ (dividing roots by k):**
    - Replace x in the old equation with kx.
8. **Roots are $\alpha^{\frac{1}{n}}$ and $\beta^{\frac{1}{n}}$:**
    - Replace x in the old equation with x^n.

Note on Proofs vs. Results

The lecturer clarifies that these are established *results* or *methods* for transforming roots, not derivations or proofs in the traditional sense. They are akin to formulas like $(a + b)^2 = a^2 + 2ab + b^2$, which are direct consequences of algebraic manipulation. While proofs can be provided, the focus in a competitive exam context is on understanding and applying these results efficiently.

Nature of Roots and the Discriminant

Introduction to Nature of Roots

The nature of the roots of a quadratic equation $ax^2 + bx + c = 0$ is determined by the discriminant, denoted by D or Δ. The formula for the roots is given by: $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$ Here, the term $b^2 - 4ac$ is the discriminant (D).

Cases for the Discriminant

1. **If $D > 0$ (Discriminant is positive):**

2. If $D = 0$ **(Discriminant is zero):**
    - The roots are **real and equal**.
    - Both roots are $x = \frac{-b}{2a}$. This is often referred to as a single repeated root.
3. If $D < 0$ **(Discriminant is negative):**
    - The roots are **complex**.
    - These complex roots are always **conjugates** of each other, *provided that the coefficients A, B, and C are real numbers*.

Important Consideration for Complex Roots

When $D < 0$, the roots are complex. However, for the roots to be complex conjugates (i.e., if one root is $p + iq$, the other is $p - iq$), a crucial condition must be met:

- **Condition for Conjugate Complex Roots:** The coefficients A, B, and C of the quadratic equation $Ax^2 + Bx + C = 0$ **must all be real numbers**.
- **Scenario where roots are not conjugate:** If at least one of the coefficients A, B, or C is a complex number, then even if $D < 0$, the roots will be complex but **not necessarily conjugates**. For example, if one root is $a + ib$, the other root might be a different complex number $c + id$, not its conjugate $a - ib$.

This distinction is very important for JEE Advanced level questions. Always check the nature of the coefficients (real or complex) before concluding that complex roots are conjugates.

Nature of Roots Based on Discriminant and Coefficients

Conditions for Rational Roots

If $Ax^2 + Bx + C = 0$, where A, B, C are rational numbers, and the discriminant $D = B^2 - 4AC$ is a perfect square and a rational number, then the roots of the quadratic equation will be rational.

Conditions for Irrational Roots

If A, B, C are rational numbers and the discriminant $D = B^2 - 4AC$ is not a perfect square (meaning its square root is irrational), then the roots of the quadratic equation will be irrational.

Conditions for Complex Conjugate Roots

If A, B, C are real numbers and one root is $P + iQ$ (a complex number), then the other root must be its conjugate, $P - iQ$. This occurs when the discriminant $D < 0$.

perfect square), then the other root must be its conjugate, $A - \sqrt{B}$. This occurs when the discriminant D is not a perfect square, and A, B, C are rational (no irrational numbers in A, B, C).

Conditions for Integer Roots

If $A = 1$, and B, C are integers, and the roots of the equation $Ax^2 + Bx + C = 0$ are rational numbers, then the roots must be integers.

Example: For the equation $x^2 - 5x + 6 = 0$, here $A = 1$, $B = -5$, and $C = 6$ are integers. The roots are 2 and 3, which are rational and also integers.

One as a Root of an Equation

Condition for One as a Root

If the sum of the coefficients of any polynomial equation is zero, then 1 is one of its roots.

For example, consider the quadratic equation $x^2 - 3x + 2 = 0$. The coefficients are 1, −3, and 2. Their sum is $1 + (-3) + 2 = 0$. Therefore, $x = 1$ is a root of this equation.

Finding Other Roots

- **For Quadratic Equations:** If the sum of the coefficients is zero, and one root is 1, the other root can be found using the product of roots formula. For a quadratic equation $ax^2 + bx + c = 0$, the product of the roots (say α and β) is given by $\alpha\beta = \frac{c}{a}$. If $\alpha = 1$, then $1 \cdot \beta = \frac{c}{a}$, which means $\beta = \frac{c}{a}$. So, for a quadratic equation, if the sum of coefficients is zero, the roots are 1 and $\frac{c}{a}$.

  Example: For $x^2 - 4x + 3 = 0$, the sum of coefficients is $1 - 4 + 3 = 0$. So, one root is 1. The other root is $\frac{c}{a} = \frac{3}{1} = 3$.

- **For Higher Degree Equations (Cubic, etc.):** If the sum of the coefficients is zero, 1 is still a root. However, finding the other roots requires further methods (e.g., synthetic division to reduce the degree of the polynomial), as the simple $\frac{c}{a}$ rule only applies to the second root of a quadratic equation.

Nature of Roots Based on Discriminant

Discriminant and Rational Roots

For a quadratic equation $ax^2 + bx + c = 0$, where a, b, c are rational numbers:

- If the discriminant $D = b^2 - 4ac$ is a perfect square, then the roots are rational.

Conjugate Roots for Irrational Cases

If a, b, c are rational numbers and the discriminant $D = b^2 - 4ac$ is not a perfect square (meaning the roots are irrational), then if one root is of the form $p + \sqrt{q}$, the other root must be its conjugate, $p - \sqrt{q}$.

Integral Roots Condition

If $a = 1$, and b and c are integers, and the discriminant $D = b^2 - 4ac$ is a perfect square integer, then the quadratic equation has integral roots.

Example: For $x^2 - 3x + 2 = 0$, $a = 1$, $b = -3$, $c = 2$. b and c are integers. The discriminant is $D = (-3)^2 - 4(1)(2) = 9 - 8 = 1$. Since 1 is a perfect square, the roots are integers. The roots are 1 and 2.

Introduction to Common Roots

Understanding Common Roots

When two quadratic equations are given, we often need to determine if they share any roots. This concept is known as 'common roots'. There are two main scenarios:

1. **Both roots are common:** This occurs when the two equations are essentially identical, meaning one is a scalar multiple of the other.
2. **Only one root is common:** This means the equations share exactly one root, while their other roots are distinct.

There is no specific condition for 'no roots common'; if the conditions for one or both common roots are not met, then there are no common roots.

Condition for Both Roots Common

Condition for Both Roots to be Common

If two quadratic equations, say:

$a_1x^2 + b_1x + c_1 = 0$

and

$a_2x^2 + b_2x + c_2 = 0$

have both roots common (i.e., $\alpha_1 = \alpha_2$ and $\beta_1 = \beta_2$), then their coefficients must be proportional. This means one equation is a scalar multiple of the other.

Example: Consider the equations:

1. $x^2 - 3x + 2 = 0$
2. $2x^2 - 6x + 4 = 0$

Here, if we compare the coefficients:

$\frac{1}{2} = \frac{-3}{-6} = \frac{2}{4}$

All ratios are equal to $\frac{1}{2}$. This indicates that the second equation is simply the first equation multiplied by 2. Therefore, both roots of these equations are common.

Condition for One Root Common

Condition for Only One Root to be Common

If two quadratic equations, $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$, have only one root common, the condition is given by:

$(c_1a_2 - c_2a_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1)$

This can be remembered using the mnemonic: $(CA)^2 = (AB)(BC)$, where:

- CA = $c_1a_2 - c_2a_1$
- AB = $a_1b_2 - a_2b_1$
- BC = $b_1c_2 - b_2c_1$

Finding the Common Root

If there is one common root, say α, it can be found using the formula:

$\alpha = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}$

This can be remembered as $\frac{CA}{AB}$.

Alternatively, to find the common root without using the formula, you can eliminate the x^2 term. If you subtract one equation (after suitable multiplication to make the x^2 coefficients equal) from the other, you will obtain a linear equation in x, which will give you the common root.

Pacing and Approach to JEE Preparation

Understanding the Pace and Depth of Teaching

- The instructor acknowledges the need to adjust the teaching pace based on student feedback.

Notes

- The instructor emphasizes that if more time is spent on detailed theory, the number of problems covered will inevitably decrease.

Common Roots Revisited

- A student requested a repeat of the concept of common roots.
- **Condition for Common Roots:** If two quadratic equations have both roots in common, then the ratio of their corresponding coefficients must be equal.
  - For two quadratic equations $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$, if both roots are common, then: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
- **Imaginary Roots:** If tw e common roots, and these roots are imagi

Did you find these notes helpful?

Manthan Batch Mindset for JEE

- The instructor stresses the importance of adopting a